

THEORY AND METHODS
TEST STATISTICS FOR DISCRIMINATING
BETWEEN AUTOREGRESSIVE MODELS OF
CONSECUTIVE ORDERS

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Summary: This paper introduces two statistics that can be used on the basis of short or moderately long series for discriminating between autoregressive processes of consecutive orders. The cases of known and unknown innovation variances are both considered. Letting Σ_p denote the covariance matrix associated with an autoregressive process of order p , the test statistics are based on a certain representation of $\Sigma_p^{-1}\Sigma_{p-1}$. A simple recursive formula for the moments of a quadratic form in normal random variables is derived, and an approximation to the density function of random quantities that are approximately distributed as gamma random variables is provided. Since quadratic forms are ubiquitous in Statistics and many statistics such as multiples of loglikelihoods have an asymptotic chi-square distribution, on their own, these two results, which are used herein to approximate the distributions of the proposed statistics, have numerous other potential applications. As shown graphically, the approximations are nearly exact and, as illustrated by two simulation studies, the proposed test statistics are outperforming Akaike's information criterion.

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1. Introduction

In time series analysis, one often wishes to test whether the correlation structure of an observed series or its residuals follows an autoregressive scheme, in which case the order of the process has to be determined. Several authors have tackled various aspects of this problem including Quenouille (1957), Akaike (1973), Shibata (1976), Ljung and Box (1978), Roy (1985), Tiao and Tsay (1989), and Pukkila et al. (1990). However, most of these results are of an asymptotic nature. The object of this paper is to develop test statistics for discriminating between two autoregressive processes of consecutive orders on the basis of short or moderately long time series, and to obtain accurate approximations to their distributions.

Box and Jenkins (1976, Section 6.2.2) refer to Anderson (1942) when discussing series of moderate length in connection with the estimated values of the autocorrelations. In the latter, a series of length $n > 75$ is considered large, in which case the exact and approximate distributions of the serial correlation are deemed to be in sufficiently close agreement, whereas it is not recommended to make use of the approximate distribution for $n < 30$, the series being too short. Accordingly, series of lengths between 30 and 75 could be characterized as having a moderate length.

We consider the following autoregressive model of order p denoted by $AR(p)$:

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \varepsilon_t \quad (1)$$

where $(\phi_1, \dots, \phi_p)$ are autoregressive parameters and the ε_t 's are independently and identically distributed normal random variables with mean 0 and variance σ^2 . The stationarity condition for the process described by (1) is that the zeros of $(1 - \phi_1 z - \dots - \phi_p z^p)$ lie outside the unit circle.

Alternatively, denoting the backward shift operator by B , one can write (1) as

$$(1 - r_1 B) \dots (1 - r_p B) X_t = \varepsilon_t \quad (2)$$

and require that $|r_i| < 1$, $i = 1, \dots, p$, to satisfy the stationary condition. Assuming that the parameters can be accurately estimated, we propose statistics for testing

$$H_0 : \text{AR}(p-1) \text{ vs. } H_1 : \text{AR}(p). \quad (3)$$

Both the cases of known and unknown innovation variances are considered, the corresponding statistics being expressed in terms of quadratic forms in Sections 2 and 3, respectively. A recursive formula for evaluating the moments of a quadratic form in normal random variables is derived in Section 4. Moment-based density approximants which apply to numerous statistics including those proposed herein are provided in Section 5. The accuracy of the approximations is verified graphically in Section 6. Two simulation studies also illustrate that the proposed statistics are quite adept at detecting the correct autoregressive order.

2. Testing $\text{AR}(p-1)$ vs. $\text{AR}(p)$ when σ^2 is known

Assuming that σ^2 is known, the ratio of the likelihood functions under the null and alternative hypotheses is

$$T_{p-1,p} = \frac{(\sigma^2)^{-n/2} (2\pi)^{-n/2} |\Sigma_{p-1}|^{-1/2} \exp \left\{ -\frac{1}{2\sigma^2} \mathbf{X}' \Sigma_{p-1}^{-1} \mathbf{X} \right\}}{(\sigma^2)^{-n/2} (2\pi)^{-n/2} |\Sigma_p|^{-1/2} \exp \left\{ -\frac{1}{2\sigma^2} \mathbf{X}' \Sigma_p^{-1} \mathbf{X} \right\}} \quad (4)$$

Under H_0 , $|\Sigma_p| = |\Sigma_{p-1}|$ and (4) then becomes

$$T_{p-1,p} = \exp \left\{ -\frac{1}{2\sigma^2} \mathbf{X}' (\Sigma_{p-1}^{-1} - \Sigma_p^{-1}) \mathbf{X} \right\}.$$

Thus

$$Y_{p-1,p} \equiv -2\sigma^2 \ln T_{p-1,p} = \mathbf{X}' (\Sigma_{p-1}^{-1} - \Sigma_p^{-1}) \mathbf{X} \quad (5)$$

can be used as a statistic for testing the null hypothesis that a series $\mathbf{X} = (X_1, \dots, X_n)'$ follows an $\text{AR}(p-1)$ process with $r_1, \dots, r_{p-1}$ as the roots of its characteristic equation, versus the alternative hypothesis that the process is $\text{AR}(p)$ with $r_1, \dots, r_p$ as the roots of its characteristic equation, Σ_d^{-1} denoting the inverse of the covariance matrix associated with an autoregressive process of order d .

Now, letting $\mathbf{X} \sim \Sigma_{p-1}^{1/2} \mathbf{Z}$ where $\Sigma_{p-1}^{1/2}$ denotes the symmetric square root of Σ_{p-1} and $\mathbf{Z} \sim \mathcal{N}_n(0, I)$, that is, $\mathbf{Z}$ is a standard normal vector, it is seen that under the null hypothesis,

$$Y_{p-1,p} \sim \mathbf{Z}' \left(I - \Sigma_{p-1}^{1/2} \Sigma_p^{-1} \Sigma_{p-1}^{1/2} \right) \mathbf{Z},$$

or, equivalently, that $Y_{p-1,p} \sim \mathbf{Z}' \Lambda \mathbf{Z}$ where Λ is a diagonal matrix whose diagonal elements are the eigenvalues of $I - \Sigma_{p-1}^{1/2} \Sigma_p^{-1} \Sigma_{p-1}^{1/2}$, which are $\lambda_i = 1 - \theta_i$, $i = 1, \dots, n$ where the θ_i 's are the eigenvalues of $\Sigma_p^{-1} \Sigma_{p-1}$. Since $Y_{p-1,p}$ is a monotonically decreasing function of the likelihood ratio statistic $T_{p-1,p}$, which can be seen from Equation (5), one would reject the null hypothesis for large values of $Y_{p-1,p}$.

Haddad (2004) showed that $\Sigma_p^{-1} \Sigma_{p-1} \left(= (\Sigma_{p-1}^{-1} \Sigma_p)^{-1} \right)$ can be expressed as $\Sigma_1^{-1}(r_p) \Delta_{p-1}^{-1}$ where

$$\Sigma_1^{-1}(r_p) = \begin{pmatrix} 1 & -r_p & 0 & \dots & 0 \\ -r_p & 1 + r_p^2 & -r_p & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & -r_p & 1 + r_p^2 & -r_p \\ 0 & \dots & 0 & -r_p & 1 \end{pmatrix}$$

and the elements of the matrix Δ_{p-1}^{-1} are given by

$$d^{jk} = \begin{cases} 1 & \text{if } j = k = 2, \dots, n-1 \\ G & \text{if } j = k = 1 \text{ or } n \\ -\left(r_p \phi_2^Y + \dots + r_p^{p-2} \phi_{p-1}^Y\right) & \text{if } j = 2 \text{ and } k = 1 \\ \vdots & \\ -r_p \phi_{p-1}^Y & \text{if } j = p-1 \text{ and } k = 1 \\ d^{(n-j+1),1} & \text{if } k = n \\ 0 & \text{elsewhere} \end{cases}$$

with

$$G = \prod_{j=1}^{p-1} (1 - r_p r_j),$$

ϕ_i^Y being the coefficient of B^i in the product $(1 - r_1 B) \dots (1 - r_{p-1} B)$, $i = 2, \dots, p-1$. Letting λ_i^+ and λ_i^- respectively denote the positive and negative eigenvalues of

$$I - \Sigma_{p-1}^{1/2} \Sigma_p^{-1} \Sigma_{p-1}^{1/2}$$

and v be the number of positive eigenvalues, one has

$$Y_{p-1,p} \sim \sum_{i=1}^v \lambda_i^+ Z_i^2 - \sum_{i=v+1}^n |\lambda_i^-| Z_i^2. \quad (6)$$

Hence $Y_{p-1,p}$ is an indefinite quadratic form whose distribution can be determined for instance by means of Imhof's (1961) integral formula or by making use of the series representation obtained by Provost and Rudiuk (1996). Alternatively, the moments and the approximate density function of each sum in (6) can be determined from Theorem 4.1 and Equation (25). Then, a simple transformation of variables yields the approximate distribution of $Y_{p-1,p}$.

3. Testing $\text{AR}(p-1)$ vs. $\text{AR}(p)$ when σ^2 is unknown

The problem of testing $H_0 : \text{AR}(p-1)$ vs. $H_1 : \text{AR}(p)$ when the innovation variance, σ^2 , is unknown, is considered in this section. In this case, the maximum likelihood estimates of σ^2 are

$$\hat{\sigma}_0^2 = \frac{\mathbf{X}'\Sigma_{p-1}^{-1}\mathbf{X}}{n} \text{ under } H_0 \quad (7)$$

and

$$\hat{\sigma}_1^2 = \frac{\mathbf{X}'\Sigma_p^{-1}\mathbf{X}}{n} \text{ under } H_1. \quad (8)$$

The likelihood ratio then becomes

$$\begin{aligned} \lambda &= \frac{\left(2\pi\hat{\sigma}_0^2\right)^{-n/2} |\Sigma_{p-1}|^{-1/2} \exp\left\{-\frac{\mathbf{X}'\Sigma_{p-1}^{-1}\mathbf{X}}{2\hat{\sigma}_0^2}\right\}}{\left(2\pi\hat{\sigma}_1^2\right)^{-n/2} |\Sigma_p|^{-1/2} \exp\left\{-\frac{\mathbf{X}'\Sigma_p^{-1}\mathbf{X}}{2\hat{\sigma}_1^2}\right\}} \\ &= \delta \left(\frac{\hat{\sigma}_1^2}{\hat{\sigma}_0^2}\right)^{n/2} \end{aligned} \quad (9)$$

where

$$\delta = \frac{|\Sigma_p|^{1/2}}{|\Sigma_{p-1}|^{1/2}}. \quad (10)$$

Thus

$$S_{p-1,p} \equiv \left(\frac{\lambda}{\delta}\right)^{2/n} = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_0^2} = \frac{\mathbf{X}'\Sigma_p^{-1}\mathbf{X}}{\mathbf{X}'\Sigma_{p-1}^{-1}\mathbf{X}} \quad (11)$$

can be used as a test statistic. This is in fact a monotonic function of the test statistic C_5 given in Choi (1992, Chapter 6). Since $S_{p-1,p}$ is a monotonically increasing function of the likelihood ratio statistic λ , which can be seen from Equation (11), one would reject the null hypothesis in favour of the alternative for small values of $S_{p-1,p}$.

Under H_0 ,

$$S_{p-1,p} \sim \frac{\mathbf{Z}' \Sigma_{p-1}^{1/2} \Sigma_p^{-1} \Sigma_{p-1}^{1/2} \mathbf{Z}}{\mathbf{Z}' \mathbf{Z}}, \quad (12)$$

while under H_1 ,

$$S_{p-1,p} \sim \frac{\mathbf{Z}' \mathbf{Z}}{\mathbf{Z}' \Sigma_p^{1/2} \Sigma_{p-1}^{-1} \Sigma_p^{1/2} \mathbf{Z}} \quad (13)$$

where $\mathbf{Z} \sim \mathcal{N}_n(0, I)$. Thus under H_0 ,

$$E(S_{p-1,p}^r) = \frac{E\left(\mathbf{Z}' \Sigma_{p-1}^{1/2} \Sigma_p^{-1} \Sigma_{p-1}^{1/2} \mathbf{Z}\right)^r}{E(W^r)}, \quad (14)$$

W denoting a chi-square random variable with n degree of freedom whose r th moment $E(W^r) = 2^r \frac{\Gamma(r + n/2)}{\Gamma(n/2)} = n(n+2) \dots (n+2r-2)$. This follows from a result stated in Hannan (1979), according to which the r th moment of the ratio of two quadratic forms is equal to the ratio of their r th moments whenever the matrix of the quadratic form appearing in the denominator is idempotent. The moments of the numerator can be determined from the recursive relationship derived in Section 4. Then, as explained in Section 5, the distribution of $S_{p-1,p}$ can be approximated in terms of Laguerre polynomials on the basis of its moments. We note that the moments and the distribution of $V_{p-1,p} \equiv 1/S_{p-1,p}$ can be similarly determined under the alternative hypothesis.

4. The moments of a quadratic form in normal random variables

A simple representation of the integer moments of quadratic forms in normal vectors is derived in this section.

By successive applications of Leibnitz's differentiation rule to the cumulant-generating function of the quadratic form $Q = \mathbf{U}' A \mathbf{U}$ where A is a

symmetric matrix and $\mathbf{U} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$, $\Sigma > 0$, that is, $\mathbf{U}$ is a p -variate normal random vector with mean vector $\boldsymbol{\mu}$ and covariance matrix Σ , Σ being positive definite, Mathai and Provost (1992, Theorem 3.2b.2) obtained the following representation of the m th moment of Q :

$$E(Q^m) = \sum_{h_1=0}^{m-1} \binom{m-1}{h_1} g^{(m-1-h_1)} \sum_{h_2=0}^{h_1-1} \binom{h_1-1}{h_2} g^{(h_1-1-h_2)} \sum_{h_3=0}^{h_2-1} \binom{h_2-1}{h_3} g^{(h_2-1-h_3)} \dots \quad (15)$$

where $g^{(k)}$ denotes the $(k+1)$ th cumulant of Q , which is defined in Equation (18) and an empty sum is interpreted as unity. (When $h_1 = m-1$, $h_2 = m-2, \dots, h_{m-1} = 1$, the right-hand side of Equation (15) contains m sums, the last sum containing the single term $g^{(0)}$.) Accordingly,

$$E(Q^{m+1}) = \sum_{h_0=0}^m \binom{m}{h_0} g^{(m-h_0)} \left\{ \sum_{h_1=0}^{h_0-1} \binom{h_0-1}{h_1} g^{(h_0-1-h_1)} \sum_{h_2=0}^{h_1-1} \binom{h_1-1}{h_2} g^{(h_1-1-h_2)} \sum_{h_3=0}^{h_2-1} \binom{h_2-1}{h_3} g^{(h_2-1-h_3)} \dots \right\}. \quad (16)$$

On substituting h_0 for m in Equation (15), the expression in brace brackets appearing on the right-hand side of Equation (16) is seen to be equal to $E(Q^{h_0})$, and the $(m+1)$ th moment of Q can then be written as

$$E(Q^{m+1}) = \sum_{h_0=0}^m \binom{m}{h_0} g^{(m-h_0)} E(Q^{h_0}).$$

This result is now stated as a theorem (with $m+1 = r$).

Theorem 4.1 Let $\mathbf{U} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$, $\Sigma > 0$, and $Q = \mathbf{U}'A\mathbf{U}$ where A is assumed to be symmetric. Then, the r th moment of the quadratic form Q

is given by

$$E(Q^r) = \sum_{i=0}^{r-1} \binom{r-1}{i} g^{(r-i-1)} E(Q^i), \quad r = 1, 2, \dots, \quad (17)$$

with $E(Q^0) = 1$, where

$$\begin{aligned} g^{(k)} &= 2^k k! \left\{ \text{tr}[(A\Sigma)^{k+1}] + (k+1) \boldsymbol{\mu}' (A\Sigma)^k A\boldsymbol{\mu} \right\} \\ &= 2^k k! \left\{ \sum_{j=1}^p \lambda_j^{k+1} + (k+1) \boldsymbol{\mu}' (A\Sigma)^k A\boldsymbol{\mu} \right\}, \quad k = 0, 1, \dots, \end{aligned} \quad (18)$$

is the $(k+1)$ th cumulant of Q and $\text{tr}[\Xi]$ denotes the trace of the matrix Ξ (which is equal to the sum of its diagonal elements), the λ 's being the eigenvalues of $A\Sigma$.

5. Density approximants expressed in terms of Laguerre polynomials

A convenient methodology for approximating the distribution of a wide array of statistics defined on the positive half-line, such as those whose distributions can be expressed exactly or asymptotically as linear combinations of chi-square random variables, is presented in this section. The proposed density approximants are expressed as the product of a gamma density function and a linear combination of Laguerre orthogonal polynomials. Series expansions based on Laguerre polynomials were used for instance by Gurland (1955) and Tiku (1965) for approximating the density functions of quadratic forms in normal variables and those of non-central χ^2 and F random variables. It is assumed that the distribution being approximated has finite moments of all orders, that it is uniquely determined by its moment sequence, and that a gamma distribution constitutes a suitable initial approximant.

Let $\mu_Y(j) = E(Y^j)$, $j = 0, 1, \dots$, denote the j th raw moment of a random variable Y defined on the interval $(0, \infty)$, which on the basis of its first two moments can be approximated by a gamma random variable with parameters $\alpha + 1$ and c , where

$$c = \frac{\mu_Y(2) - \mu(1)^2}{\mu_Y(1)} \quad (19)$$

and

$$\alpha = \frac{\mu_Y(1)}{c} - 1. \quad (20)$$

Denoting the j th moment of $X = Y/c$ by $\mu_X(j) = E[(Y/c)^j] = \mu_Y(j)/c^j$, the density function of the random variable X , which is also defined on the interval $(0, \infty)$, can be approximated by

$$f_{X_d}(x) = \sum_{j=0}^d a_j \varphi_j(x) \quad (21)$$

where

$$\varphi_j(x) = x^\alpha e^{-x} \mathcal{L}_j^{(\alpha)}(x) \quad (22)$$

and

$$a_j = (-1)^j \sum_{i=0}^j (-1)^i \frac{j!}{j!(j-i)!\Gamma(\alpha+j-i+1)} \mu_X(j-i), \quad (23)$$

$\mathcal{L}_j^{(\alpha)}(x)$ denoting a Laguerre polynomial of degree j with parameter $\alpha > -1$, which can be expressed as follows:

$$\mathcal{L}_j^{(\alpha)}(x) = (-1)^j \sum_{i=0}^j \frac{(-1)^i \Gamma(\alpha+j+1)}{i!(j-i)!\Gamma(\alpha+j-i+1)} x^{j-i}, \quad j = 0, 1, \dots \quad (24)$$

For alternative representations, related results and convergence properties in connection with the truncated orthogonal series appearing in Equation (21), the reader is referred to Szegö (1959), Abramowitz and Stegun (1970) and Provost

(2005). The density approximant for $Y = cX$ is thus given by

$$f_{Y_d}(y) = \frac{y^\alpha e^{-y/c}}{\Gamma(\alpha+1) c^{\alpha+1}} \sum_{j=0}^d \Gamma(\alpha+1) a_j \mathcal{L}_j^{(\alpha)}(y/c), \quad y > 0. \quad (25)$$

6. Simulation studies

Assuming that σ^2 is known, $n = 30$ and $p = 3$, 100000 replications of the quadratic form $Y_{2,3} = \mathbf{Z}'(I - \Sigma_2^{1/2}\Sigma_3^{-1}\Sigma_2^{1/2})\mathbf{Z}$ where $\mathbf{Z} \sim \mathcal{N}_{30}(0, I)$ were simulated. The cumulative distribution function of $Y_{2,3}$, as evaluated by integration of its density approximant, is superimposed on the simulated cumulative distribution function in Figure 1. In this case, the approximate density function was obtained on the basis of six moments by applying Theorem 4.1 and Equation (25) to each sum appearing in the representation of $Y_{p-1,p}$ given in Equation (6), as was explained in Section 2. Similarly, the statistic $S_{2,3} = \mathbf{Z}'\Sigma_2^{1/2}\Sigma_3^{-1}\Sigma_2^{1/2}\mathbf{Z}/(\mathbf{Z}'\mathbf{Z})$ was simulated under the assumption that σ^2 is unknown. Its moments and approximate distribution were also determined from the results derived in Sections 3 and 4. The cumulative distribution function of $S_{2,3}$, which was obtained from a sixth-degree Laguerre polynomial density approximation, is superimposed on the simulated cumulative distribution function in Figure 2. The approximate distribution functions are seen to be in close agreement with their simulated counterparts, whether σ^2 is assumed to be known or unknown.

Two simulation studies were carried out to ascertain the extent to which each of the test statistics is adept at discriminating between two autoregressive processes of successive orders. First, one hundred series of lengths 20 and 30 were generated from white noise, and $H_0 : \text{AR}(0)$, that is, $\Sigma_0 = I$, was tested against $H_1 : \text{AR}(1)$. The frequencies out of 100 corresponding to the correct identification of the white noise process at the 5% significance level,

which were obtained by making use of the proposed statistics and Akaike's information criterion, are reported in Table 1. The results indicate that the proposed statistics can outperform the AIC.

Table 1. $H_0 : \text{AR}(0)$ vs. $H_1 : \text{AR}(1)$ under White Noise

n	Tests	# Times AR(0) Selected
20	$Y_{p-1,p}$	86
20	$S_{p-1,p}$	88
20	AIC	74
30	$Y_{p-1,p}$	96
30	$S_{p-1,p}$	92
30	AIC	90

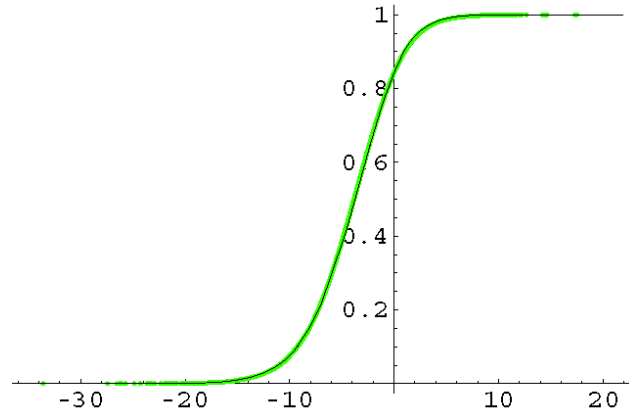


Figure 1. Approximate CDF of $Y_{2,3}$ (solid line) superimposed on the simulated CDF

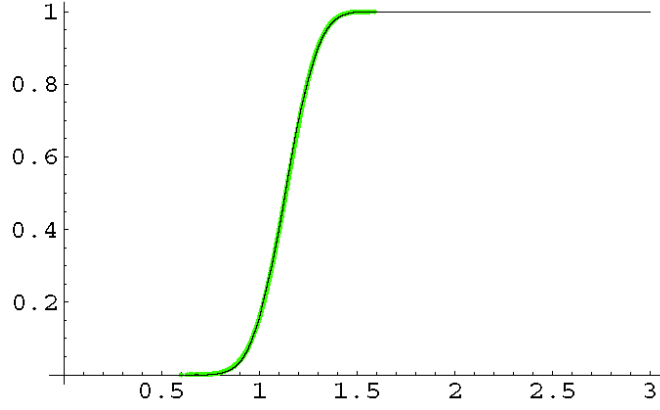


Figure 2. Approximate CDF of $S_{2,3}$ (solid line) superimposed on the simulated CDF

Secondly, one hundred series of lengths 20 and 30 were generated from the second-order autoregressive process,

$$X_t - 0.3X_{t-1} - 0.3X_{t-2} = \varepsilon_t, \quad (26)$$

and the null hypothesis that a series is $AR(2)$ was tested against $H_1: AR(3)$; then, making use of the same $AR(2)$ series, the null hypothesis that a series is $AR(1)$ was tested against the alternative hypothesis that it is indeed a second-order autoregressive series. The purpose of this study was to determine whether a series of a given order would be identified as such with respect to tests of hypotheses involving lower and higher orders. The frequencies out of 100 corresponding to the correct identification of the $AR(2)$ model at the 5% significance level, which were obtained by making use of the statistics $Y_{p-1,p}$ and $S_{p-1,p}$ and the AIC, are reported in Tables 2 and 3. Overall, the proposed statistics are seen to be more adept at identifying the correct order of the underlying process than Akaike's information criterion.

Table 2. $H_0 : \text{AR}(2)$ vs. $H_1 : \text{AR}(3)$ under $\text{AR}(2)$

n	Tests	# Times $\text{AR}(2)$ Selected
20	$Y_{p-1,p}$	93
20	$S_{p-1,p}$	94
20	AIC	88
30	$Y_{p-1,p}$	94
30	$S_{p-1,p}$	92
30	AIC	85

Table 3. $H_0 : \text{AR}(1)$ vs. $H_1 : \text{AR}(2)$ under $\text{AR}(2)$

n	Tests	# Times $\text{AR}(2)$ Selected
20	$Y_{p-1,p}$	80
20	$S_{p-1,p}$	82
20	AIC	74
30	$Y_{p-1,p}$	88
30	$S_{p-1,p}$	89
30	AIC	82

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